

Reviewer 1

Comments

Reviewer 1 - Initial submission, 08/09/2026

This survey examines the applicability and fairness of BBR congestion control across three versions and six network scenarios. Its proposed contribution is a framework organized around buffer depth, bandwidth-delay product (BDP), and loss source. It also reviews intra-protocol and inter-protocol fairness and discusses opportunities for AI-assisted congestion control.

The manuscript argues that BBR performs particularly well in high-BDP environments, that its advantages depend strongly on network conditions, and that persistent fairness problems arise from structural properties of its measurement model.

Major Comments

- 1) Sections 5–5.4 and Figures 5–6 infer that a larger propagation RTT necessarily produces a larger bandwidth share. This inference is incomplete. A larger amount of in-flight data can simply be necessary to sustain the same throughput over a longer path. The product identity alone does not determine bandwidth allocation.
- 2) Table 3, Table 9, and Sections 5.2 and 6.1 imply that BBRv2/v3 fairness mechanisms fail or revert toward BBRv1 when ECN is unavailable. This overlooks the loss-driven mechanisms also described in the manuscript.
- 3) Sections 3.2–3.4 propose rules for deciding when BBR is suitable, but do not adequately explain how the numerical boundaries were established. The categories are also inconsistent: Section 3.2 defines medium buffers as $1.5\text{--}10\times\text{BDP}$, while Table 5 uses $1\text{--}10\times\text{BDP}$.
- 4) Please explain what “suitable” means: higher throughput, lower delay, fairer bandwidth sharing, fewer retransmissions, or some combination. Specify which RTT is used to calculate BDP when competing flows have different RTTs. Also explain how BBR version, queue management, and the number of flows affect the proposed rules. Provide the bandwidth and RTT values supporting the classification of each scenario, particularly LEO satellite links.
- 5) Section 1.2 and Table 1 should explain what this paper adds beyond existing surveys, especially reference [20]. Covering more scenarios may be useful, but the authors should identify the new insights gained from bringing those scenarios together.
- 6) Sections 6.2–6.3 make broad claims that most AI approaches optimize only one objective and that none addresses both fairness and different service requirements. Please provide evidence for these statements or narrow them to the studies reviewed. Also distinguish limitations of particular simulation studies from limitations of the simulation software itself.

Minor Comments

- 1) Replace rhetorical formulations such as “This is a geometric fact, not a bug,” “nemeses,” and “operators have no reason” with precise, qualified scientific language.
- 2) Reduce repetition across the abstract, framework, scenario matrix, fairness discussion, and conclusion.
- 3) In Figure 2, the delay that increases with queue occupancy should not be labeled propagation delay.

Reviewer 2

Comments

Reviewer 2 - Initial submission, 15/09/2026

This manuscript provides a systematic review of the applicability boundaries, version evolution, fairness issues, and AI-assisted optimization of BBR congestion control. It develops a three-dimensional applicability framework based on buffer depth, BDP magnitude, and loss source, and comprehensively analyzes the performance and fairness of BBRv1–v3 across multiple representative network scenarios. The manuscript further discusses RTT fairness, the coexistence of BBR and CUBIC, and potential directions for structural improvement. Overall, the topic is practically relevant and the coverage is broad. However, the following issues remain:

(1) The systematic review process and the 165-item evidence set lack reproducibility. Although Figure 1 outlines the search and screening stages and reports the final inclusion of 165 sources with some information on their composition, it does not provide the key counts needed to reconstruct the selection process. These include the numbers of records retrieved from each database, the numbers before and after deduplication, the numbers screened and excluded during title/abstract screening, and the numbers assessed and excluded during full-text review. More importantly, the manuscript explicitly treats “the 165 body-text citations” as the final statistical unit while also listing “genuinely cited in the text” as an inclusion criterion. This creates a clear circular criterion: references are included because they are cited in the main text, while the final 165 cited references are then used to demonstrate the systematic nature of the review. Based on the current reporting, readers cannot determine whether these 165 sources were genuinely derived from a predefined and reproducible search and screening process or were retrospectively assembled from the reference set after the manuscript had been completed. The authors should provide complete search and screening numbers, deduplication rules, and reasons for full-text exclusion. They should also clearly explain how the final 165-item evidence set was progressively derived from the initial search results.

(2) The three-dimensional framework lacks explicit decision criteria and adequate validation. The manuscript defines buffer depth, BDP magnitude, and loss source as three “orthogonal” physical dimensions. It also claims that the framework is quantifiable, falsifiable, and applicable to algorithm selection. However, several key problems remain. First, the High, Medium, and Low “suitability” ratings in Table 4 do not have clear and consistent criteria. The manuscript does not explain whether these ratings are based on a single metric, such as throughput, latency, or fairness, or on thresholds combining multiple metrics. Consequently, readers cannot independently reproduce these ratings or use them to make consistent algorithm-selection decisions. Second, Section 3.5 acknowledges that the three dimensions are not strictly orthogonal, that loss source cannot be accurately represented as a static attribute, and that flow count has not been

incorporated into the framework. These limitations indicate that the relationships among the dimensions and their effects on the ratings have not been sufficiently explained, and the reliability of the framework for algorithm selection requires further validation. Third, the “independent validation” in Section 3.5 mainly uses the data from Scholz et al. [25] to conduct qualitative consistency checks for P1–P3. This is closer to showing that the external data do not clearly contradict the framework than to validating its predictive ability using data that were not involved in constructing the rules.

(3) Table 3 contains errors in its summary of the key parameters and mechanisms of BBRv1, BBRv2, and BBRv3. First, the target inflight for BBRv2/v3 during ProbeRTT is not simply set to $BDP/2$. Instead, it is the larger of 4 MSS and $BDP/2$. Second, `cwnd_gain` is not always fixed at 2.0 across BBRv1–v3. Third, the note below the table states that BBRv2/v3 degrade toward BBRv1 when ECN support is unavailable. This description is misleading. Even without ECN marking, BBRv2/v3 retain loss-feedback-based mechanisms for updating the `bw_lo`, `inflight_lo`, and `inflight_hi` bounds, together with the corresponding state-control logic. Therefore, the absence of ECN alone is insufficient to conclude that the overall behavior of BBRv2/v3 approaches that of BBRv1.

(4) The preparation of Figures 3–6 and the extent of AI involvement require further clarification. The existing “Declaration of generative AI and AI-assisted technologies” mentions figure-text translation but does not explicitly state whether generative AI was used to generate these figures. If AI-assisted generation was involved, the authors should clearly disclose the tools used, their specific purposes, and the extent of their use in both the declaration and the corresponding figure captions. The authors should also explain how they verified the accuracy of the technical content presented in these figures.

Reviewer 3

Comments

Reviewer 3 - Initial submission, 17/09/2026

The authors have empirically evaluated the various versions of BBR. The results discussion is well presented. The drawback of this current version of the paper is as follows:

1. The abstract is not easy to follow. It should be revised and written as per the style of the Abstract. Remove the reference content, reduce the numerical detail, and focus on the motivation, objective, and contribution in a nutshell.
2. The authors have focused on only BBR variants. They should also discuss other BDP-based TCPs, such as BDP-Veno and OVENO. They should increase the literature part.
3. The authors have carried out only the numerical evaluation. The proposed algorithm or analytical study is missing, thereby limiting the paper's novelty. The author should also contribute in these terms.
4. The implementation details are not sufficient. It must be defined in detail, including the tools, platform, and configuration. That may help the reader reproduce this research work.
5. The rest of the presentation related to the study on BBR is well done.